POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6043C | Semester VI | Theory | 4 Credits

Electronics Product Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6043C as Electronics Product Engineering in Semester VI for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Roorkee’s Product Design and Development course and polytechnic product engineering curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Product designs, PCB layouts, manufacturing recipes and certification tests must be based on approved engineering plans, certified hardware data, current industry standards and competent professional review.

Verified source note: Verified sources support product life cycles, PCB design, DFM, and EMC concepts. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Electronics Product Engineering studies the systematic process of designing, developing and manufacturing electronic products. It develops the ability to analyze product life cycles, evaluate

PCB design and fabrication techniques, understand EMI/EMC and thermal management, and coordinate product engineering solutions while respecting the technical and industrial boundaries of electronic and manufacturing engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□□□ syllabus topic □□□□□□ □□□□□□□□□□□; □□□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□□ □□□□□□□□□□.

Prerequisite foundation

Basic electronics, circuit theory, electronic components and basic manufacturing processes.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the stages of the electronic product design and development process.
2. Analyze the principles of PCB design, layout and fabrication for electronic circuits.
3. Evaluate the systematic design for manufacturing (DFM) and electromagnetic compatibility (EMC).
4. Explain the testing, prototyping and documentation requirements for electronic products.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the electronic product design life cycle and development stages.	Understand
CO2	Analyze PCB design rules, layout techniques and fabrication processes.	Apply
CO3	Evaluate design for manufacturing (DFM) and EMI/EMC considerations.	Apply
CO4	Explain prototyping, testing and documentation for electronic products.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Product Design Life Cycle and Development	Introduction to product engineering; Product life cycle (PLC); Design stages: Concept, Feasibility, Design, Prototyping, Production; Market research and user requirements.	Approx. 14 hours	CO1	Module 1
2	PCB Design and Fabrication	PCB types: Single, Double, and Multi-layer; Schematic capture; Component selection and footprints; PCB layout rules: Trace width, Clearance, Via types; Fabrication process: Etching, Drilling, Soldermask.	Approx. 16 hours	CO2	Module 2
3	DFM, Thermal Management and EMI/EMC	Design for Manufacturing (DFM) and Assembly (DFA); Thermal management: Heat sinks, Fan cooling; EMI/EMC: Sources of interference, Shielding, Grounding techniques.	Approx. 16 hours	CO3	Module 3
4	Prototyping, Testing and Documentation	Prototyping techniques: Breadboarding, Perfboarding, 3D printing for enclosures; Testing: Functional, Stress, and Environmental testing; Product documentation: BOM, Assembly drawings.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Product Design Life Cycle and Development

Introduction to product engineering; Product life cycle (PLC); Design stages: Concept, Feasibility, Design, Prototyping, Production; Market research and user requirements.

CO1 · Approx. 14 hours

PCB Design and Fabrication

PCB types: Single, Double, and Multi-layer; Schematic capture; Component selection and footprints; PCB layout rules: Trace width, Clearance, Via types; Fabrication process: Etching, Drilling, Soldermask.

CO2 · Approx. 16 hours

DFM, Thermal Management and EMI/EMC

Design for Manufacturing (DFM) and Assembly (DFA); Thermal management: Heat sinks, Fan cooling; EMI/EMC: Sources of interference, Shielding, Grounding techniques.

CO3 · Approx. 16 hours

Prototyping, Testing and Documentation

Prototyping techniques: Breadboarding, Perfboarding, 3D printing for enclosures; Testing: Functional, Stress, and Environmental testing; Product documentation: BOM, Assembly drawings.

CO4 · Approx. 14 hours

MODULE 1

Product Design Life Cycle and Development

Introduction to product engineering; Product life cycle (PLC); Design stages: Concept, Feasibility, Design, Prototyping, Production; Market research and user requirements.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

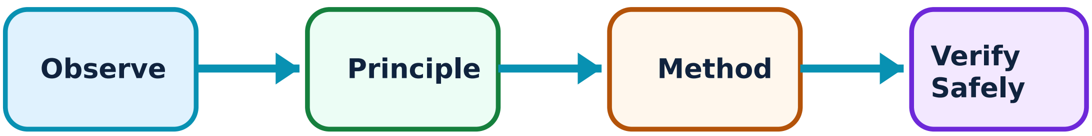
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Product Design Life Cycle and Development: identify the system, state its principle, follow the correct method and verify the result safely.

1. Product Engineering Fundamentals−

Product engineering is the process of translating a concept into a manufacturable product. The 'Product Life Cycle' (PLC) includes introduction, growth, maturity, and decline. Systematic design ensures that the product meets user needs, quality standards, and cost targets.

Study and application method

List the stages of the 'Electronic Product Design' process; explain the significance of the 'Maturity' stage in the PLC.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the stages of the 'Electronic Product Design' process; explain the significance of the 'Maturity' stage in the PLC.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Ignoring user requirements can lead to product failure. Do not bypass the feasibility study without authorised market analysis and technical risk assessment.

2. Design Specifications−

Specifications define the technical and functional requirements of the product. This includes electrical parameters, mechanical dimensions, environmental constraints, and regulatory compliance (e.g., RoHS, CE).

Study and application method

Describe the difference between 'Functional' and 'Technical' specifications; list three 'Environmental' constraints for electronic products.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the difference between 'Functional' and 'Technical' specifications; list three 'Environmental' constraints for electronic products.

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Common mistake / precaution: Incomplete specifications lead to design revisions. All specifications must follow authorised review and sign-off by all stakeholders before detailed design begins.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

PCB Design and Fabrication

PCB types: Single, Double, and Multi-layer; Schematic capture; Component selection and footprints; PCB layout rules: Trace width, Clearance, Via types; Fabrication process: Etching, Drilling, Soldermask.

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

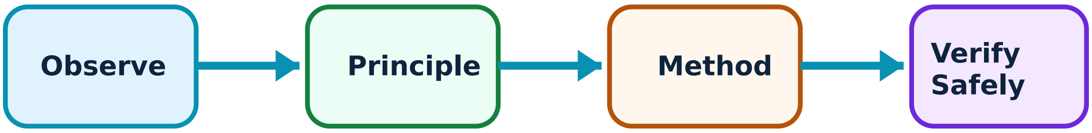
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for PCB Design and Fabrication: identify the system, state its principle, follow the correct method and verify the result safely.

1. PCB Layout Principles−

PCB (Printed Circuit Board) design involves placing components and routing traces. Key rules include maintaining minimum trace width for current, clearance for insulation, and using appropriate via types for layer transitions. High-speed designs require controlled impedance.

Study and application method

Sketch a 'Double-sided PCB' stack-up; explain the purpose of a 'Ground Plane' in PCB design.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a 'Double-sided PCB' stack-up; explain the purpose of a 'Ground Plane' in PCB design.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper trace width can cause overheating. All PCB layouts must follow authorised current-carrying capacity charts and clearance standards for safety.

2. PCB Fabrication and Assembly−

Fabrication involves transferring the layout to the board using etching, followed by drilling and applying soldermask. Assembly (PCBA) involves soldering components using Through-hole or Surface Mount Technology (SMT).

Study and application method

Describe the steps of the 'Chemical Etching' process; compare 'Through-hole' and 'SMT' assembly conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the steps of the 'Chemical Etching' process; compare 'Through-hole' and 'SMT' assembly conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Soldering defects (cold joints, bridges) can cause intermittent failures. All assembly must follow authorised IPC standards for soldering and inspection.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

DFM, Thermal Management and EMI/EMC

Design for Manufacturing (DFM) and Assembly (DFA); Thermal management: Heat sinks, Fan cooling; EMI/EMC: Sources of interference, Shielding, Grounding techniques.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

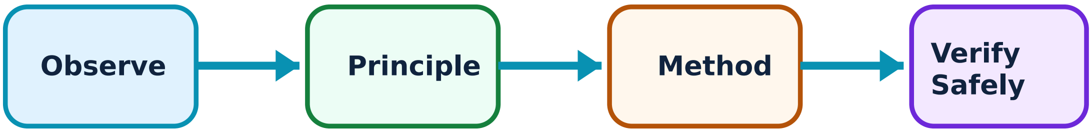
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for DFM, Thermal Management and EMI/EMC: identify the system, state its principle, follow the correct method and verify the result safely.

1. Design for Excellence (DFX)−

DFM (Design for Manufacturing) ensures the product is easy and cost-effective to make. DFA (Design for Assembly) focuses on reducing assembly time and errors. These concepts are critical for mass production.

Study and application method

Explain the concept of 'DFM' in electronic product design; list two benefits of 'DFA'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the concept of 'DFM' in electronic product design; list two benefits of 'DFA'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Ignoring manufacturing constraints increases cost. Do not finalize designs without authorised 'Design Rule Checks' (DRC) and manufacturing review.

2. Thermal and Electromagnetic Integrity−

Electronic components generate heat that must be dissipated using heat sinks or fans. EMI (Electromagnetic Interference) can disrupt circuit operation. EMC (Electromagnetic Compatibility) is achieved through proper grounding, decoupling, and shielding.

Study and application method

Describe the role of a 'Heat Sink' in thermal management; explain how 'Shielding' reduces EMI conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the role of a 'Heat Sink' in thermal management; explain how 'Shielding' reduces EMI conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Poor thermal management leads to premature component failure. All high-power designs must follow authorised thermal simulation and worst-case temperature analysis.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Prototyping, Testing and Documentation

Prototyping techniques: Breadboarding, Perfboarding, 3D printing for enclosures; Testing: Functional, Stress, and Environmental testing; Product documentation: BOM, Assembly drawings.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

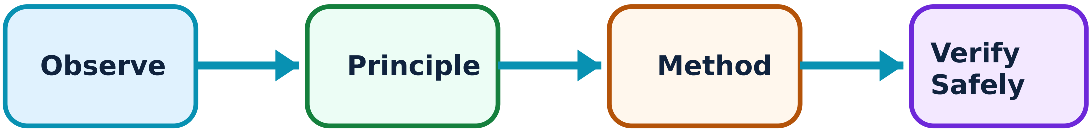
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Prototyping, Testing and Documentation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Prototyping and Testing−

Prototyping validates the design before mass production. Functional testing checks if the product works. Stress testing evaluates performance under extreme conditions. Environmental testing (heat, humidity) ensures reliability in the field.

Study and application method

Differentiate between 'Functional' and 'Stress' testing; describe the role of '3D Printing' in enclosure prototyping.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Functional' and 'Stress' testing; describe the role of '3D Printing' in enclosure prototyping.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Prototype behavior may differ from production units. Do not rely solely on breadboard results for high-frequency or high-power verification without authorised PCB prototyping.

2. Documentation and Certification−

Comprehensive documentation is essential for manufacturing and maintenance. This includes the Bill of Materials (BOM), schematics, and assembly drawings. Regulatory certification (CE, UL, FCC) is required for market entry.

Study and application method

Explain the purpose of a 'Bill of Materials' (BOM); list three 'Regulatory Certifications' for electronic products.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the purpose of a 'Bill of Materials' (BOM); list three 'Regulatory Certifications' for electronic products.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Inaccurate documentation leads to manufacturing errors. All production files must follow authorised version control and engineering change order (ECO) procedures.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Product Design Life Cycle and Development

Use the concept flow to revise observation, principle, method and verification.

Module 2: PCB Design and Fabrication

Use the concept flow to revise observation, principle, method and verification.

Module 3: DFM, Thermal Management and EMI/EMC

Use the concept flow to revise observation, principle, method and verification.

Module 4: Prototyping, Testing and Documentation

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Develop a 'Bill of Materials' (BOM) conceptually for a simple regulated power supply.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a PCB layout for a 'Bridge Rectifier' circuit showing trace routing.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the thermal path from a power transistor to the ambient air showing the heat sink.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the required trace width conceptually for a 2A current using a standard PCB chart.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify an electronic product (e.g., smartphone, calculator) and note its enclosure and PCB features.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Product Design Life Cycle and Development

Explain Product Engineering Fundamentals and state one correct method, application or precaution.−

Product engineering is the process of translating a concept into a manufacturable product. The 'Product Life Cycle' (PLC) includes introduction, growth, maturity, and decline. Systematic design ensures that the product meets user needs, quality standards, and cost targets.

Answer framework: List the stages of the 'Electronic Product Design' process; explain the significance of the 'Maturity' stage in the PLC.

Important point: Ignoring user requirements can lead to product failure. Do not bypass the feasibility study without authorised market analysis and technical risk assessment.

Explain Design Specifications and state one correct method, application or precaution.−

Specifications define the technical and functional requirements of the product. This includes electrical parameters, mechanical dimensions, environmental constraints, and regulatory compliance (e.g., RoHS, CE).

Answer framework: Describe the difference between 'Functional' and 'Technical' specifications; list three 'Environmental' constraints for electronic products.

Important point: Incomplete specifications lead to design revisions. All specifications must follow authorised review and sign-off by all stakeholders before detailed design begins.

Module 2 — PCB Design and Fabrication

Explain PCB Layout Principles and state one correct method, application or precaution.−

PCB (Printed Circuit Board) design involves placing components and routing traces. Key rules include maintaining minimum trace width for current, clearance for insulation, and using appropriate via types for layer transitions. High-speed designs require controlled impedance.

Answer framework: Sketch a 'Double-sided PCB' stack-up; explain the purpose of a 'Ground Plane' in PCB design.

Important point: Improper trace width can cause overheating. All PCB layouts must follow authorised current-carrying capacity charts and clearance standards for safety.

Explain PCB Fabrication and Assembly and state one correct method, application or precaution.−

Fabrication involves transferring the layout to the board using etching, followed by drilling and applying soldermask. Assembly (PCBA) involves soldering components using Through-hole or Surface Mount Technology (SMT).

Answer framework: Describe the steps of the 'Chemical Etching' process; compare 'Through-hole' and 'SMT' assembly conceptually.

Important point: Soldering defects (cold joints, bridges) can cause intermittent failures. All assembly must follow authorised IPC standards for soldering and inspection.

Module 3 — DFM, Thermal Management and EMI/EMC

Explain Design for Excellence (DFX) and state one correct method, application or precaution.−

DFM (Design for Manufacturing) ensures the product is easy and cost-effective to make. DFA (Design for Assembly) focuses on reducing assembly time and errors. These concepts are critical for mass production.

Answer framework: Explain the concept of 'DFM' in electronic product design; list two benefits of 'DFA'.

Important point: Ignoring manufacturing constraints increases cost. Do not finalize designs without authorised 'Design Rule Checks' (DRC) and manufacturing review.

Explain Thermal and Electromagnetic Integrity and state one correct method, application or precaution.−

Electronic components generate heat that must be dissipated using heat sinks or fans. EMI (Electromagnetic Interference) can disrupt circuit operation. EMC (Electromagnetic Compatibility) is achieved through proper grounding, decoupling, and shielding.

Answer framework: Describe the role of a 'Heat Sink' in thermal management; explain how 'Shielding' reduces EMI conceptually.

Important point: Poor thermal management leads to premature component failure. All high-power designs must follow authorised thermal simulation and worst-case temperature analysis.

Module 4 — Prototyping, Testing and Documentation

Explain Prototyping and Testing and state one correct method, application or precaution.–

Prototyping validates the design before mass production. Functional testing checks if the product works. Stress testing evaluates performance under extreme conditions. Environmental testing (heat, humidity) ensures reliability in the field.

Answer framework: Differentiate between 'Functional' and 'Stress' testing; describe the role of '3D Printing' in enclosure prototyping.

Important point: Prototype behavior may differ from production units. Do not rely solely on breadboard results for high-frequency or high-power verification without authorised PCB prototyping.

Explain Documentation and Certification and state one correct method, application or precaution.–

Comprehensive documentation is essential for manufacturing and maintenance. This includes the Bill of Materials (BOM), schematics, and assembly drawings. Regulatory certification (CE, UL, FCC) is required for market entry.

Answer framework: Explain the purpose of a 'Bill of Materials' (BOM); list three 'Regulatory Certifications' for electronic products.

Important point: Inaccurate documentation leads to manufacturing errors. All production files must follow authorised version control and engineering change order (ECO) procedures.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Product Design Life Cycle and Development.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: PCB Design and Fabrication.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: DFM, Thermal Management and EMI/EMC.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Prototyping, Testing and Documentation.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Introduction to product engineering; Product life cycle (PLC); Design stages: Concept, Feasibility, Design, Prototyping, Production; Market research and user requirements..
2. Develop a complete answer or practical plan for: PCB types: Single, Double, and Multi-layer; Schematic capture; Component selection and footprints; PCB layout rules: Trace width, Clearance, Via types; Fabrication process: Etching, Drilling, Soldermask..
3. Develop a complete answer or practical plan for: Design for Manufacturing (DFM) and Assembly (DFA); Thermal management: Heat sinks, Fan cooling; EMI/EMC: Sources of interference, Shielding, Grounding techniques..
4. Develop a complete answer or practical plan for: Prototyping techniques: Breadboarding, Perfboarding, 3D printing for enclosures; Testing: Functional, Stress, and Environmental testing; Product documentation: BOM, Assembly drawings..

LAST-MILE REVISION

Quick Revision

Module 1: Product Design Life Cycle and Development

Introduction to product engineering; Product life cycle (PLC); Design stages: Concept, Feasibility, Design, Prototyping, Production; Market research and user requirements.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: PCB Design and Fabrication

PCB types: Single, Double, and Multi-layer; Schematic capture; Component selection and footprints; PCB layout rules: Trace width, Clearance, Via types; Fabrication process: Etching, Drilling, Soldermask.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: DFM, Thermal Management and EMI/EMC

Design for Manufacturing (DFM) and Assembly (DFA); Thermal management: Heat sinks, Fan cooling; EMI/EMC: Sources of interference, Shielding, Grounding techniques.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Prototyping, Testing and Documentation

Prototyping techniques: Breadboarding, Perfboarding, 3D printing for enclosures; Testing: Functional, Stress, and Environmental testing; Product documentation: BOM, Assembly drawings.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Product Engineering Fundamentals	Product engineering is the process of translating a concept into a manufacturable product.
Design Specifications	Specifications define the technical and functional requirements of the product.
PCB Layout Principles	PCB (Printed Circuit Board) design involves placing components and routing traces.
PCB Fabrication and Assembly	Fabrication involves transferring the layout to the board using etching, followed by drilling and applying soldermask.
Design for Excellence (DFX)	DFM (Design for Manufacturing) ensures the product is easy and cost-effective to make.
Thermal and Electromagnetic Integrity	Electronic components generate heat that must be dissipated using heat sinks or fans.
Prototyping and Testing	Prototyping validates the design before mass production.
Documentation and Certification	Comprehensive documentation is essential for manufacturing and maintenance.

LEARNING RESOURCES

References

Recommended electronics product engineering references

1. C. H. Mangin and S. McClelland, Surface Mount Technology: The Future of Electronics Assembly.
2. R. S. Khandpur, Printed Circuit Boards: Design, Fabrication, Assembly and Testing.
3. V. Prasad, Electronic Product Design.
4. Mark I. Montrose, EMC and the Printed Circuit Board: Design, Theory, and Layout Made Simple.

Approved public electronics product engineering sources

- https://nptel.ac.in/courses/112107217
- https://onlinecourses.nptel.ac.in/noc21_me66/preview

These sources provide the verified study basis while official Course 6043C detail is unavailable; they do not replace official industrial designs, authorised safety certifications, certified equipment specifications or authorised strategic product plans.